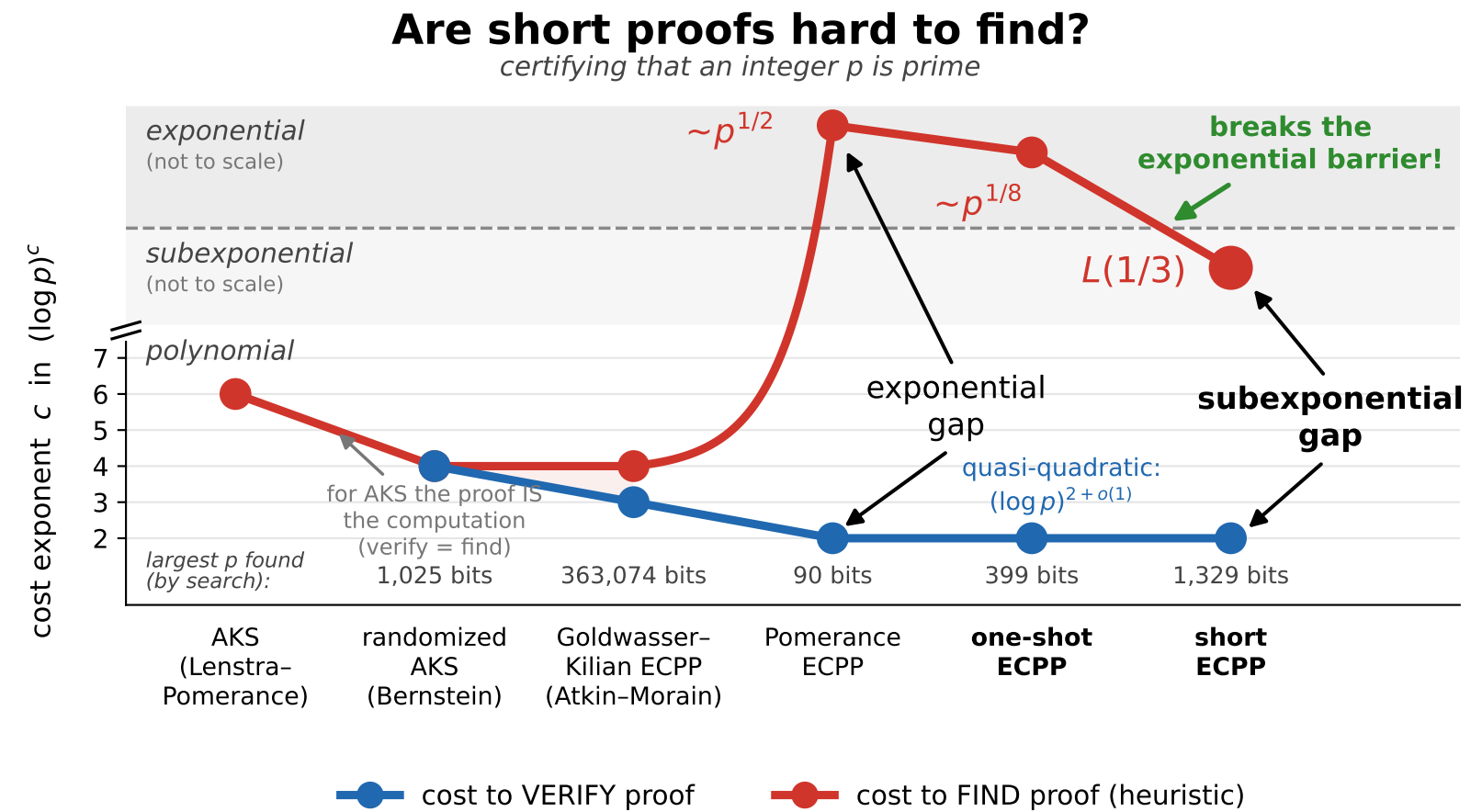
Are short proofs hard to find?
certifying that an integer p is prime
exponential
(not to scale)
subexponential
(not to scale)
polynomial
cost exponent c in (log p)^c
7
6
5
4
3
2
~p^1/2
~p^1/8
L(1/3)
breaks the exponential barrier!
exponential gap
subexponential gap
quasi-quadratic: (log p)^2 + o(1)
for AKS the proof IS the computation (verify = find)
largest p found (by search):
1,025 bits
363,074 bits
90 bits
399 bits
1,329 bits
AKS (Lenstra–Pomerance)
randomized AKS (Bernstein)
Goldwasser–Kilian ECPP (Atkin–Morain)
Pomerance ECPP
one-shot ECPP
short ECPP
cost to VERIFY proof
cost to FIND proof (heuristic)